NORMAL VECTOR ESTIMATION

A research internship report

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Agenda

03	Introdution
04	Literature review
05	Method Explanation
06	Deployment & Usage

01 Focused on improving algorithms from geometrical aspects to enhance accuracy and robustness.

02 Conducted a comprehensive literature review to understand existing approaches and identify gaps.

03 Designed and integrated improvements to address limitations.



Overview

- Strengthened understanding of research methodology and critical analysis.
- Gained insights into how theory translates into practical algorithm design and evaluation.

Skills and tools gained

in this section , we will discuss everything we gained across the internship.

Technical skills

- Strengthened understanding of computer vision, geometry, and optimization concepts.
- Applied data analysis and experimental validation techniques

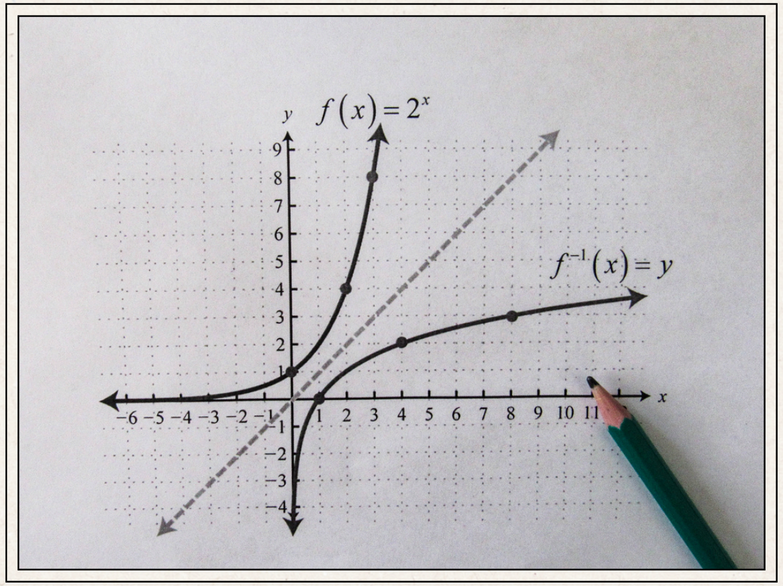
Research

- Conducted systematic literature reviews and synthesized findings.
- Learned to bridge theory with practical implementation.

Tools

- python
- c++
- deployment





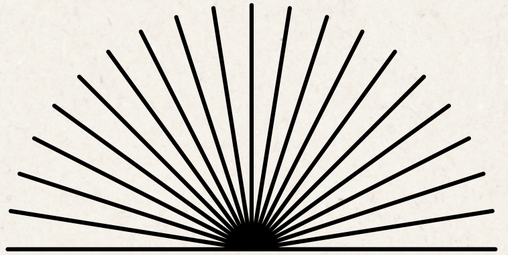
Method

- Purpose: Estimate surface orientation at each point in a 3D point cloud. Essential for tasks like surface reconstruction, feature extraction, and registration.
- Method:

1. Select a neighborhood around each point (e.g., k-nearest neighbors).
2. Apply Principal Component Analysis (PCA) to the covariance matrix of the neighborhood.
3. The eigenvector corresponding to the smallest eigenvalue is the estimated normal vector.

- Challenges:
- Choosing the right neighborhood size (too small → noisy, too large → loss of detail).
- Orientation ambiguity (normals may flip; need consistency checks).
- Applications: Widely used in computer vision, robotics, and 3D modeling.

Thank you



- I would like to express my sincere gratitude for your time and attention throughout this presentation.
- This internship has been a valuable opportunity that allowed me to grow both technically and personally.
- I am deeply thankful to my supervisors and colleagues for their guidance, support, and encouragement during this journey.
- Finally, I appreciate the chance to share my work with you today, and I welcome any questions or feedback you may have.

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